

Maps in Dart

A Map stores data as key-value pairs.

```
Map<String, int> scores = { 'Math': 90, 'Science': 85, };
```

Adding & Updating Values

```
scores['English'] = 88; scores['Math'] = 95;
```

Removing Values

```
scores.remove('Science');
```

Accessing Map Data

```
print(scores['Math']);
```

Looping Through a Map

```
scores.forEach((key, value) { print('$key : $value'); });
```

Null Safety in Dart

Null safety helps prevent errors caused by accessing null values.

Benefits of Null Safety

- Fewer runtime errors
- Clearer and safer code
- Better IDE support

Nullable vs Non-Nullable

```
String name = "Rana"; // Non-nullable String? middleName; // Nullable
```

Null Assertion Operator (!)

```
print(middleName!.length);
```



Use with caution: It throws an error if the value is null.

Late Keyword

```
late String description; description = "Dart Language";
```



Control Flow Statements

Control flow determines how the program executes.

Conditional Statements

```
int number = 10; if (number > 0) { print("Positive"); } else if (number == 0) { print("Zero"); } else { print("Negative"); }
```

Switch Case

```
int day = 3; switch (day) { case 1: print("Monday"); break; case 2: print("Tuesday"); break; default: print("Invalid"); }
```


Loops in Dart

Loops are used to repeat code.



For Loop

```
for (int i = 1; i <= 5; i++) { print(i); }
```

While Loop

```
int i = 1; while (i <= 5) { print(i); i++; }
```

Do-While Loop

```
int i = 1; do { print(i); i++; } while (i <= 5);
```